CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 1: Interdisciplinary Computing & Systems Modeling

Course: Multidisciplinary Minor-III [23CSMDXX]
Category: Assignments | Assignment #1 | Academic Year: 2026-27

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1. Overview & Objective:

Mathematical modeling, dynamical systems representation, and interdisciplinary computational problem framing.

2. Assignment Problems & Instructions:

1. Formulate state-space representations of physical and biological systems.
2. Apply numerical simulation methods Euler, Runge-Kutta 4th Order to differential models.
3. Analyze system feedback loops, steady-state equilibria, and sensitivity indices.
4. Construct computational abstractions for cross-domain engineering problems.

Submission Guidelines & Verification:

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